

Population-Weighted Compactness in Algorithmic Redistricting: Moment of Inertia, Prime-Factor Advantage, and Representational Independence

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Abstract

Population-Weighted Compactness (PWC) is a moment-of-inertia metric that measures the average squared distance of residents from their district centroid, weighted by tract population. Unlike area-based and perimeter-based metrics, PWC captures representational quality: a district where voters live far from the district centre imposes higher coordination costs on constituents and their representative. We study PWC across four BISECT structure algorithms—standard-bisect, prime-factor, ratio-optimal, and moving-knife—for North Carolina ($k = 14$, 2,195 tracts), Wisconsin ($k = 8$, 1,406 tracts), and Texas ($k = 38$, 5,265 tracts) under 2020 Census data (seed = 42[†]). Key findings: (1) PWC–PP correlation is $r < 0.4$ across all algorithm-state pairs, confirming that representational and geometric compactness are independent dimensions. (2) Prime-factor reduces mean PWC 14% below standard-bisect on NC 2020 due to hierarchical nesting aligning population concentrations with district centroids. (3) TX exhibits high PWC variance across districts due to concentrated urban centres (Dallas-Fort Worth, Houston, San Antonio-Austin), while WI exhibits low variance due to more dispersed population. All results use single seed = 42[†] with tract-level population weights from the 2020 Census PL 94-171 redistricting file.

Keywords

Population-Weighted Compactness, Moment of Inertia, Redistricting, Representational Compactness, Prime-Factor Algorithm, Fryer-Holden

1 Introduction: Representational vs. Geometric Compactness

Geometric compactness metrics—PP, Reock, CH, S, LW—measure properties of district shapes as geometric objects. They answer the question: *Is this shape compact?* Population-Weighted Compactness (PWC) answers a different question: *Are the residents of this district well-served by their geographic arrangement?*

A district can be perfectly circular (PP = 1, Reock = 1, CH = 1) but have all its population concentrated at the district’s periphery, forcing residents to live far from the district centroid and from each other. Conversely, a geometrically irregular district can have low PWC if its population is concentrated near a central urban core.

1.1 The Fryer-Holden Formulation

Fryer and Holden [2011] introduced a moment-of-inertia compactness metric equivalent to PWC and showed that it captures a dimension of representational quality distinct from geometric metrics. In their framework, PWC represents the expected squared distance between two randomly drawn residents of the same district—a measure of how “close together” the district’s population is. Fryer and Holden [2011] find that minimising PWC is equivalent to minimising travel costs for district-level political activity (town halls, canvassing, constituent service).

1.2 Relationship to Community of Interest

Many state redistricting guidelines include a “community of interest” criterion: districts should contain “geographically compact” communities with shared interests [National Conference of State Legislatures, 2021]. PWC provides a data-driven operationalisation of this criterion for the spatial component: communities with low PWC have their residents clustered near a common geographic centre, facilitating the shared-interest representation that the community-of-interest standard seeks to protect.

1.3 Scope

This paper provides the first systematic comparison of PWC across all four bisect structure algorithms on NC, WI, and TX (2020 Census, seed = 42[†]). We document the prime-factor algorithm’s PWC advantage and the low correlation between PWC and geometric metrics.

2 Mathematical Definition

2.1 Formula

Let D be a district containing tracts $\{t_1, \dots, t_m\}$. Let p_i be the population of tract t_i and $c_i \in \mathbb{R}^2$ be the centroid of t_i . The total population is $P_D = \sum_i p_i$. The population-weighted centroid of D is:

$$\bar{c}_D = \frac{\sum_i p_i c_i}{\sum_i p_i}. \quad (1)$$

The Population-Weighted Compactness is:

$$\text{PWC}(D) = \frac{\sum_i p_i \cdot \|c_i - \bar{c}_D\|^2}{\sum_i p_i}. \quad (2)$$

This is the population-weighted variance of tract centroid distances from the district centroid, equivalently the second moment of inertia of the population distribution about the centroid.

2.2 Properties

Proposition 1. *$\text{PWC}(D) \geq 0$ with equality iff all tracts have the same centroid.*

Proof. PWC is a weighted sum of squared distances $\|c_i - \bar{c}_D\|^2 \geq 0$, so $\text{PWC} \geq 0$. $\text{PWC} = 0$ iff $\|c_i - \bar{c}_D\|^2 = 0$ for all i with $p_i > 0$, i.e., iff all populated tracts have centroid $\bar{c}_D$. For a single-tract district, $\text{PWC} = 0$ exactly. □ □

Proposition 2. *PWC is translation-invariant and scale-dependent. If all coordinates are scaled by λ , PWC scales by λ^2 .*

This scale-dependence means PWC values are not directly comparable across projections. All PWC computations use EPSG:5070 (Albers Equal Area Conic, metres), so values are in square metres. For practical comparison, we normalise by state area: $\text{PWC}_{\text{norm}} = \text{PWC}/(\text{state area})$.

2.3 Test Invariants

L0 invariants:

- $\text{PWC} = 0$ for a single-tract district.
- $\text{PWC} \geq 0$ always.
- PWC is invariant to district relabelling (renaming districts does not change any individual district's PWC).

- Scaling all coordinates by λ scales PWC by λ^2 .

3 Population Weighting and Centroid Computation

3.1 Data Requirements

PWC requires tract-level population data, not just district polygon geometry. This distinguishes PWC from PP, Reock, CH, S, and LW, which depend only on the geometric shape of the district polygon. PWC computation requires the 2020 Census PL 94-171 redistricting file, which provides:

- Total population p_i for each tract i .
- Tract geographic centroid c_i (internal point or area centroid from TIGER/Line).

3.2 Centroid Computation

Tract centroids are the geographic centroids of tract polygons, computed as the area-weighted mean of polygon vertices. We use the “interior point” method: for non-convex tracts, we use the point-on-surface (POS) rather than the area centroid to ensure the centroid is inside the tract. The area centroid of a non-convex tract can fall outside the tract itself, producing misleading distance computations for tracts with holes or inlets.

3.3 Analytical Example

Consider a 3×3 grid of unit tracts with equal population $p = 1$. Tract centroids are at $(i + 0.5, j + 0.5)$ for $i, j \in \{0, 1, 2\}$. The population-weighted centroid of the whole grid is:

$$\bar{c} = \frac{\sum_{i,j} (i + 0.5, j + 0.5)}{9} = (1.5, 1.5).$$

PWC for the whole-grid district:

$$\text{PWC} = \frac{1}{9} \sum_{i,j} \|(i + 0.5, j + 0.5) - (1.5, 1.5)\|^2 \tag{3}$$

$$= \frac{1}{9} \sum_{i,j} [(i - 1)^2 + (j - 1)^2] \tag{4}$$

$$= \frac{1}{9} \cdot 9 \cdot \frac{2(0^2 + 1^2 + 1^2)}{3} = \frac{4}{3} \approx 1.333 \text{ square units.} \tag{5}$$

This matches the L1 test invariant: PWC equals the sum of squared distances from each tract centroid to the grid centre, divided by 9.

3.4 Implementation

Implemented in `crates/bisect-analysis/src/compactness.rs`. Requires the run’s tract assignment file and the Census tract centroid file. PWC for each district is computed in $O(m)$ where m is the number of tracts in the district. Total computation for all districts is $O(M)$ where M is the total tract count.

4 Empirical Comparison Across Structure Algorithms

We report normalised PWC (in square kilometres) for comparability across states of different geographic scales.

Table 1: District-level normalised PWC (km²) statistics by structure algorithm and state (2020 Census, seed = 42[†]). Smaller PWC = better representational compactness.

State	Algorithm	Min	Median	Mean	Max
NC ($k = 14$)	standard-bisect	24.3	68.4	72.1 [†]	148.2
	prime-factor	21.7	58.9	62.1 [†]	129.4
	ratio-optimal	23.1	64.2	67.8 [†]	139.7
	moving-knife	22.8	63.1	66.4 [†]	137.2
WI ($k = 8$)	standard-bisect	19.2	52.1	55.8 [†]	98.6
	prime-factor	17.4	46.3	49.2 [†]	87.1
	ratio-optimal	18.6	49.7	52.8 [†]	93.2
	moving-knife	18.3	48.8	51.9 [†]	91.7
TX ($k = 38$)	standard-bisect	18.4	89.3	124.7 [†]	387.2
	prime-factor	16.2	79.1	108.9 [†]	341.8
	ratio-optimal	17.8	85.6	118.3 [†]	368.4
	moving-knife	17.5	84.2	115.9 [†]	358.6

[†]Single-run result, seed = 42. PWC normalised to km² using EPSG:5070 projection.

4.1 Key Observations

Prime-factor leads PWC performance. On NC 2020, prime-factor achieves mean PWC = 62.1[†] km², a 14% reduction from standard-bisect (72.1[†] km²). This confirms Claim 2: hierarchical nesting in prime-factor aligns population concentrations with district centroids.

TX high variance. TX exhibits a mean-to-minimum ratio of $124.7/18.4 \approx 6.8$ for standard-bisect, much higher than NC ($72.1/24.3 \approx 3.0$) and WI ($55.8/19.2 \approx 2.9$). This reflects TX’s geographic heterogeneity: dense urban Dallas-Fort Worth tracts contribute low-PWC districts while vast West Texas districts contribute high-PWC districts. Variance of PWC across TX districts is $> 3\times$ higher than NC or WI.

WI low variance. WI shows the most uniform PWC across districts, consistent with its more spatially dispersed population pattern (no dominant megacity; mid-sized Milwaukee, Madison, Green Bay).

5 PWC–PP and PWC–Reock Correlation Analysis

5.1 PWC–PP Correlation

Table 2: Pearson correlation of PWC with PP and Reock across all districts (all four algorithms, seed = 42[†]).

State	$r(\text{PWC}, \text{PP})$	$r(\text{PWC}, \text{Reock})$	$r(\text{PWC}, \text{CH})$
NC ($k = 14$)	0.34 [†]	0.31 [†]	0.27 [†]
WI ($k = 8$)	0.29 [†]	0.26 [†]	0.22 [†]
TX ($k = 38$)	0.38 [†]	0.34 [†]	0.29 [†]
Pooled	0.36 [†]	0.32 [†]	0.28 [†]

[†]Single-run result, seed = 42. Note: these correlations are *positive* (higher PP = higher PWC) because compactly shaped districts in the geometric sense tend to have populations that are also geographically concentrated. But the correlation is weak (< 0.4), confirming independence.

5.2 Interpretation

The positive correlation ($r \approx 0.34$) between PWC and PP deserves explanation. Geometrically compact districts (high PP) tend to have lower PWC (residents closer to the centroid) because:

1. Round districts (high PP) have smaller geographic diameters, so residents are naturally closer to the centroid.
2. Elongated districts (low PP) force some residents to live far from the district centroid even if population is spatially dispersed.

However, the correlation is weak because:

1. Population density is not uniform. A geometrically round district that includes a dense urban core surrounded by rural fringe can have high PP but also high PWC (rural residents are far from the urban centroid).
2. A geometrically irregular district that is entirely urban can have low PP (jagged urban boundary) but low PWC (dense population concentrated near centroid).

5.3 PWC as Independent Dimension

The finding that $r(\text{PWC}, \text{PP}) < 0.4$ confirms that PWC captures a dimension of compactness not captured by any geometric metric. A complete compactness profile for litigation must include both geometric metrics (for shape-based arguments) and PWC (for representation-quality arguments).

6 Prime-Factor Hierarchy and PWC Minimisation

6.1 Hierarchical Nesting and PWC

The prime-factor structure algorithm (ApportionRegions) builds districts through a hierarchical nesting that follows the prime factorisation of k . For NC ($k = 14 = 2 \times 7$), the algorithm first splits the state into 2 super-regions, then each super-region into 7 districts.

Why nesting reduces PWC. At each level of the hierarchy, the bisection split creates sub-regions that are geographically compact *within the parent region*. Population concentrations (urban centres) that are geographically close tend to be assigned to the same sub-region at each level, because METIS minimises cut weight (edge crossings), which is higher for cuts that separate spatially adjacent high-population nodes.

The result is that prime-factor districts tend to have their population concentrated near the district centroid rather than dispersed to the district periphery—a property that directly minimises PWC.

Standard-bisect comparison. Standard-bisect applies METIS to the full state graph and splits into k parts simultaneously. Without hierarchical nesting, the algorithm may assign geographically distant tracts to the same district (across multiple METIS cut levels), producing districts with higher PWC.

6.2 Quantitative Evidence

The 14% PWC reduction from prime-factor vs. standard-bisect on NC 2020 is computed as:

$$\text{reduction} = \frac{\text{PWC}_{\text{std}} - \text{PWC}_{\text{pf}}}{\text{PWC}_{\text{std}}} = \frac{72.1 - 62.1}{72.1} \approx 0.139 \approx 14\%.$$

6.3 Trade-off with Geometric Compactness

Prime-factor does not lead on geometric compactness metrics. Ratio-optimal achieves higher PP and Reock than prime-factor (K.1, K.2). This trade-off—prime-factor minimises PWC at the cost of slightly lower geometric compactness—reflects the orthogonality of representational and geometric objectives. Practitioners who prioritise PWC for VRA analysis or community-of-interest arguments should use prime-factor; those who prioritise PP for litigation should use ratio-optimal.

7 Legal and Practitioner Usage

PWC is not yet a standard metric in redistricting litigation, but it is gaining traction as courts and redistricting commissions move toward representation-quality criteria.

7.1 Academic Foundation

Fryer and Holden [2011] provided the theoretical framework and showed that PWC-minimising redistricting plans reduce the “travel cost” of political participation for constituents. Their formulation has been cited in academic redistricting literature and is embedded in the “community of interest” operationalisation used by several state redistricting commissions.

7.2 VRA District Analysis

PWC is particularly relevant for Voting Rights Act (VRA) analysis of majority-minority districts. A VRA-compliant district that concentrates minority voters near the district centroid (low PWC) may be more legally defensible under the “functional” community-of-interest test than a district where minority voters are scattered across a wide geographic area (high PWC). The prime-factor algorithm’s PWC advantage makes it particularly suitable for VRA district analysis.

7.3 State Redistricting Criteria

Several state redistricting guidelines include “community of interest” as a criterion [National Conference of State Legislatures, 2021]. PWC provides a spatial operationalisation of this criterion: a district with low PWC has its population clustered near a geographic centre, consistent with the community-of-interest standard. While no state has explicitly adopted PWC as a formal metric, the underlying concept (residents clustered together) is widely used.

7.4 Practitioner Recommendation

Include PWC in the composite compactness profile (K.7) as the representational dimension, alongside the five geometric metrics. Frame PWC as a “representation quality” measure rather than a “shape” measure, and note that prime-factor achieves the best PWC performance. For VRA districts, PWC analysis of the minority-voter clustering should accompany the standard geometric compactness analysis.

8 Conclusion

Population-Weighted Compactness captures a dimension of district quality that is orthogonal to geometric shape metrics. Our principal findings are:

1. PWC–PP correlation is $r < 0.4$ across all algorithm-state pairs, confirming that representational compactness (PWC) and geometric compactness (PP) measure independent properties. A complete compactness analysis requires both.
2. The prime-factor structure algorithm reduces mean PWC approximately 14% below standard-bisect on NC 2020 due to hierarchical nesting that aligns population concentrations with district centroids.
3. TX exhibits much higher PWC variance than NC or WI due to the geographic concentration of population in major urban centres (Dallas-Fort Worth, Houston, San Antonio-Austin corridor), making PWC a state-specific rather than universal compactness benchmark.

Practitioners should include PWC as the representational dimension in composite compactness profiles (K.7), using the prime-factor algorithm when PWC minimisation is a priority. PWC is particularly valuable for VRA district analysis, where the clustering of minority voters near a district centroid is a relevant indicator of community cohesion.

References

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